

Transformations of $f(x) = |x|$, Part I - Practice

Algebra II Honors | Unit 2 | Lesson 2 Practice | TEK A2.6.C

Objective: I can prove when two transformed absolute value functions produce identical graphs, and factor combined a/b expressions into a single real a-value.

Section 1 — Recall

1. Prove algebraically that $f(x) = 3|x|$ and $f(x) = |3x|$ produce identical graphs.

2. Factor $f(x) = |6x|$ into the form $k|x|$ and give k .

$$|6x| = |6||x| \rightarrow k = \underline{\hspace{2cm}}$$

Section 2 — Apply

3. Describe every transformation of $f(x) = -(3/2)|4x|$, first factoring out the constant to find the real a-value.

$$-(3/2)|4x| = -(3/2)(4)|x| = \underline{\hspace{2cm}}$$

4. Functions $g(x) = a|x|$ and $h(x) = |bx|$ produce the SAME graph for positive a and b . Express the relationship between a and b , and justify.

$$a = \underline{\hspace{2cm}} \text{ (since } |bx| = |b||x| = b|x| \text{ when } b > 0)$$

Section 3 — Challenge

5. $f(x) = |x|$ is transformed with a vertical stretch by 5 AND reflected over BOTH the x-axis and the y-axis. Write the resulting function, and explain why one reflection has no visible effect.

$$g(x) = \underline{\hspace{2cm}}$$

6. Prove that for ANY real $b \neq 0$ (positive or negative), $f(x) = |bx|$ can always be rewritten as $k|x|$, and give k in terms of b .

$$|bx| = |b||x| \rightarrow k = \underline{\hspace{2cm}} \text{ (always non-negative, regardless of the sign of } b)$$

Section 4 — Explain It

7. Explain why every function of the form $f(x) = |bx|$ can always be rewritten in the simpler form $k|x|$, where $k = |b|$. Use a specific example to illustrate.
